

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2319

46Th Street 138 kV
September 24, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2319. The Interconnection Customer proposes a 75.6 MW solar photovoltaic facility interconnecting at the 46Th Street 138 kV substation in the New Jersey area. The request was received on June 10, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 570 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$63,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Belleville – Artesian 138 KV Line	White Rock-Nike – Magnolia 1708 138 KV	105.1	275	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Belleville – Artesian 138 KV	Reconductor line	\$678,000	9% (\$63,000)
	Total allocated to project		\$63,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.